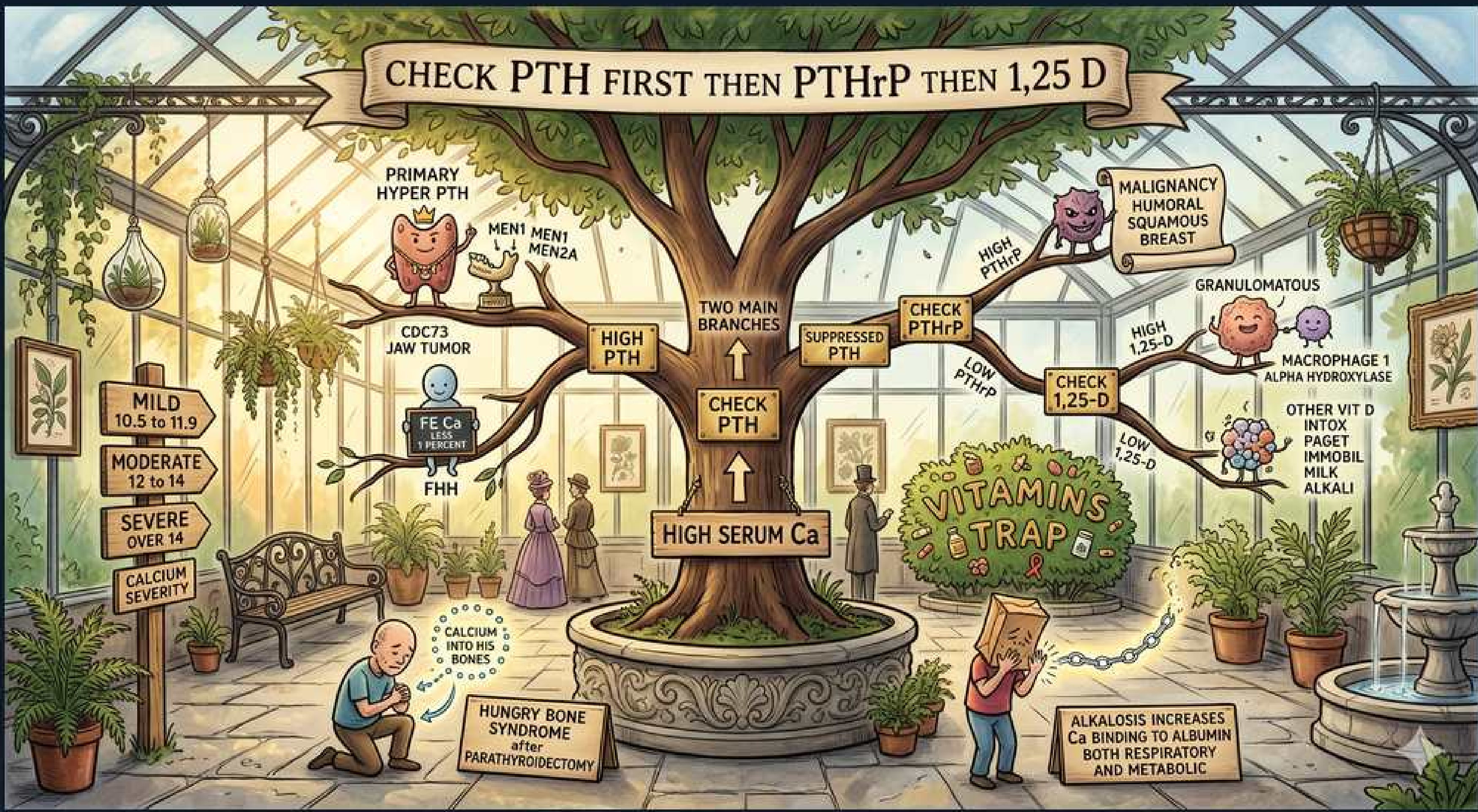
Hypercalcemia — PTH-First Diagnostic Algorithm



1. Check PTH first

WHAT YOU SEE

Tree trunk 'CHECK PTH', header 'CHECK PTH FIRST THEN PTHrP THEN 1,25 D'

WHAT IT ENCODES

For any hypercalcemia, measure PTH first. This splits causes into PTH-dependent (high/inappropriately normal PTH) vs PTH-independent (suppressed PTH). It is the single most important branch point.

BOARD TRAP

Don't order PTHrP or vitamin D metabolites before PTH — PTH dictates the entire workup direction.

2. High serum Ca = start

WHAT YOU SEE

Planter 'HIGH SERUM Ca' at tree base

WHAT IT ENCODES

Confirm true hypercalcemia first: correct calcium for albumin (add 0.8 mg/dL per 1 g/dL albumin below 4) or check ionized calcium. Pseudohypercalcemia from high albumin/dehydration is excluded.

BOARD TRAP

A high total calcium with low albumin may be normal ionized calcium — always correct for albumin.

3. High PTH branch

WHAT YOU SEE

Left branch 'HIGH PTH'

WHAT IT ENCODES

Elevated or inappropriately normal PTH with high calcium = PTH-mediated hypercalcemia: primary hyperparathyroidism, FHH, lithium, or tertiary hyperparathyroidism.

BOARD TRAP

An 'inappropriately normal' PTH (not frankly high) with hypercalcemia still indicates a PTH-driven cause — normal is abnormal here.

4. Primary hyperparathyroidism

WHAT YOU SEE

Parathyroid gland icon 'PRIMARY HYPER PTH', 'MEN1 MEN2A'

WHAT IT ENCODES

Most common cause of outpatient hypercalcemia. Usually a single adenoma; high PTH, high Ca, LOW phosphate, high urine Ca. Associated with MEN1 and MEN2A.

BOARD TRAP

Distinguish from FHH using urine calcium — primary HPT has HIGH urinary Ca; FHH has LOW urinary Ca.

5. CDC73 / jaw tumor syndrome

WHAT YOU SEE

'CDC73 JAW TUMOR'

WHAT IT ENCODES

Hyperparathyroidism–jaw tumor syndrome (CDC73/HRPT2 mutation) carries higher risk of parathyroid CARCINOMA and ossifying jaw fibromas.

BOARD TRAP

Parathyroid carcinoma causes very high Ca and PTH with a palpable neck mass — consider CDC73 in young/severe cases.

6. FHH — low urine Ca

WHAT YOU SEE

'FHH', 'LIFE Ca IN BONES'

WHAT IT ENCODES

Familial hypocalciuric hypercalcemia: inactivating CaSR mutation → mildly high Ca, high-normal PTH, and characteristically LOW urinary calcium (FE-Ca <1%). Benign, no surgery needed.

BOARD TRAP

FECa <0.01 distinguishes FHH from primary HPT — operating on FHH is a classic mistake.

7. Suppressed PTH branch

WHAT YOU SEE

Middle branch 'SUPPRESSED PTH → CHECK PTHrP'

WHAT IT ENCODES

Low/suppressed PTH with hypercalcemia = PTH-independent: malignancy, vitamin D excess, granulomatous disease, high bone turnover, milk-alkali. Next step is PTHrP and vitamin D metabolites.

BOARD TRAP

Suppressed PTH should redirect you toward malignancy/vitamin D causes — not parathyroid surgery.

8. PTHrP — humoral malignancy

WHAT YOU SEE

'MALIGNANCY HUMORAL SQUAMOUS BREAST', 'HIGH PTHrP'

WHAT IT ENCODES

Humoral hypercalcemia of malignancy: tumor-secreted PTHrP (squamous cell lung/head-neck, renal, breast). High PTHrP, suppressed PTH, low phosphate; rapid severe hypercalcemia.

BOARD TRAP

PTHrP mimics PTH biochemically but PTH assay reads LOW — measure PTHrP separately.

9. 1,25-D high — granulomatous

WHAT YOU SEE
'GRANULOMATOUS', 'MACROPHAGE 1-ALPHA HYDROXYLASE', 'CHECK 1,25-D'

WHAT IT ENCODES
Sarcoid/TB/lymphoma macrophages express 1-alpha-hydroxylase, converting 25-D to active 1,25-D → increased gut Ca absorption. High 1,25-D, suppressed PTH.

BOARD TRAP
Granulomatous hypercalcemia is driven by 1,25-D (calcitriol), responds to glucocorticoids — not by PTH or 25-D excess.

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10. 25-D high — vitamin D intox

WHAT YOU SEE
'OTHER VIT D INTOX, PAGET, IMMOBILE, MILK ALKALI', 'LOW 1,25-D'

WHAT IT ENCODES
Exogenous vitamin D toxicity raises 25-D. Other PTH-independent causes: Paget (immobilized), prolonged immobilization, milk-alkali syndrome, thiazides, vitamin A, hyperthyroidism.

BOARD TRAP
Vitamin D intoxication raises 25-D, whereas granulomatous disease raises 1,25-D — check the right metabolite.

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11. Hungry bone syndrome

WHAT YOU SEE
Kneeling figure 'HUNGRY BONE SYNDROME after PARATHYROIDECTOMY', 'CALCIUM INTO BONES'

WHAT IT ENCODES
After parathyroidectomy for severe HPT, rapid remineralization causes profound HYPOcalcemia, hypophosphatemia, hypomagnesemia. Needs aggressive calcium/vitamin D replacement.

BOARD TRAP
Post-op hypocalcemia from hungry bone is distinct from transient hypoparathyroidism — both follow neck surgery.

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12. Calcium severity bands

WHAT YOU SEE
Signpost 'MILD 10.5–11.9, MODERATE 12–14, SEVERE >14', 'CALCIUM SEVERITY'

WHAT IT ENCODES
Severity guides urgency: mild <12 (hydrate, treat cause), moderate 12–14, severe >14 or symptomatic = emergency (IV saline + calcitonin + bisphosphonate/denosumab).

BOARD TRAP
Symptoms and rate of rise matter more than absolute number — acute 13 can be worse than chronic 13.

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13. Vitamins trap (albumin/pH)

WHAT YOU SEE
'VITAMINS TRAP', 'ALKALOSIS INCREASES Ca BINDING TO ALBUMIN'

WHAT IT ENCODES
Acid-base affects ionized calcium: alkalosis increases Ca binding to albumin → lower ionized Ca (tetany of hyperventilation); acidosis frees ionized Ca.

BOARD TRAP
In alkalosis a normal TOTAL calcium can mask LOW ionized calcium causing tetany — check ionized Ca.

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14. Check PTHrP step

WHAT YOU SEE
Mid-branch 'CHECK PTHrP'

WHAT IT ENCODES
When PTH is suppressed, the next branch is PTHrP (malignancy) then vitamin D metabolites — sequential, not simultaneous, to localize the cause efficiently.

BOARD TRAP
Following the ordered sequence PTH→PTHrP→vitamin D prevents missing humoral malignancy.
